ZHANHONG HE

Music and Audio Machine Learning Researcher

zhanh.he@research.uwa.edu.au

https://github.com/zhanh-he/ | https://www.linkedin.com/in/zhanhong-he/

Research Profile

Music and audio machine learning researcher working on expressive musical dynamics, audio-symbolic modelling and music generation. Research includes MIDI velocity estimation, score-informed refinement, multitask learning and perceptual evaluation. Industrial research experience at Dolby and Huawei complements university teaching and project co-supervision. Interested in tools that give musicians understandable control over musical expression.

Education

University of Western Australia

PhD research in Electrical and Electronic Engineering | February 2022-present

Doctoral defense completed; graduation certificate expected November 2026. Formal award date pending confirmation.

Research focus: musical dynamics, audio and symbolic music processing, and machine learning.

University International Fee Scholarship; University Postgraduate Awards; 2026 UWA Data Institute Award.

University of Western Australia

BSc (Honours) in Engineering Science | October 2018-December 2021

First Class Honours; GPA 7/7; WAM 87.7/100.

Honours thesis: Improving Spectrogram-based Music Source Separation using Phase Information.

Selected Research

Expressive dynamics across instruments

- Developed research on adapting a piano-pretrained MIDI velocity estimator to other instruments using differentiable SoundFont proxies and audio-loudness supervision.
- Investigated supervision that reduces reliance on note-level velocity labels in non-piano recordings.

Score-informed velocity refinement

- Developed a Transformer correction approach using score onset and frame information to refine velocity estimates from piano transcription models.

Joint dynamics and musical structure modelling

- Developed a multitask CRNN for piano dynamic markings, beats, downbeats and structural changes.
- Reimplemented a Bark-scale specific-loudness feature extractor in PyTorch for musical audio analysis.

MIDI expression generation and listening evaluation

- Adapted a U-Net image-colourisation approach to predict per-note MIDI velocity.
- Designed a listening test conducted with support from professional listeners at Dolby.

Research Experience

Huawei Hongkong Research Centre - Research Intern, full time

Appointment commencing June 2026; six-month appointment

- Conducted research on music-model post-training and human-preference evaluation.

Dolby Laboratories, Sydney - Research Intern, full time | May-September 2024

- Developed music-generation models using U-Net, VAE and masked Transformer architectures.
- Designed and conducted professional listening tests to evaluate generated musical expression.

Great Bay University - Visiting Student Researcher, part time

Appointment commencing March 2026; three-month appointment

- Explored pretrained music models, including MERT, and agentic workflows for music information retrieval research.

Teaching And Collaboration

University of Western Australia - Tutor and Lab Facilitator, part time | February 2023-December 2025

- Supported ELEC4404 Signal Processing and GENG2000 Engineering Practices.
- Co-supervised CITS4011 Computer Science Honours projects and final-year designs.
- Collaborated with researchers across music, signal processing and machine learning through the projects and publications listed here.

Selected Publications And Preprints

Zhanhong He, Hanyu Meng, Defeng (David) Huang and Roberto Togneri. Joint Estimation of Piano Dynamics and Metrical Structure with a Multi-Task Multi-Scale Network. IEEE ICASSP, 2026. DOI: 10.1109/ICASSP55912.2026.11460839.

Zhanhong He, Hanyu Meng, David Defeng Huang and Roberto Togneri. Beyond Piano: Cross-Instrument MIDI Velocity Estimation via Differentiable SoundFont Proxies. Preprint, arXiv:2608.08985, 2026.

Zhanhong He, David Cooper, Defeng Huang and Roberto Togneri. Filling MIDI Velocity using U-Net Image Colorizer. Preprint, arXiv:2508.07751, 2025.

Music Practice And Interactive Systems

- UWA Conservatorium of Music coursework in electronic music, music generation and ensemble performance, 2023; composition and mixing with Ableton Live, and participation in a public ensemble concert.
- Built a coursework HMM system for piano and environmental-sound generation using Max/MSP with Ableton Live.
- Built an Arduino-based theremin with a MIDI interface, programmed in C, at a Physical Interaction Design for Music workshop, 2023.

Technical Skills

Python and PyTorch; CRNN, Transformer, U-Net and VAE research; audio and symbolic music processing; MIDI; perceptual evaluation; Ableton Live and Max/MSP.

Languages: English (fluent); Chinese and Cantonese (native).